Result 8.7

Three distinct points A, B and C with position vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are collinear if and only if there exist real numbers x,y,z, none of them is zero, such that

$$x+y+z=0 \text{ and } x\vec{a}+y\vec{b}+z\vec{c}=\vec{0}.$$

Example 8.3

Let A and B be two points with position vectors $2\vec{a}+4\vec{b}$ and $2\vec{a}-8\vec{b}$. Find the position vectors of the points which divide the line segment joining A and B in the ratio 1:3 internally and externally.

Solution

Let O be the origin. It is given that

$$\overrightarrow{OA}=2\vec{a}+4\vec{b} \text{ and } \overrightarrow{OB}=2\vec{a}-8\vec{b}.$$

Let C and D be the points which divide the segment AB in the ratio 1 : 3 internally and externally respectively. Then

$$\overrightarrow{OC}=\frac{3\overrightarrow{OA}+\overrightarrow{OB}}{3+1}=\frac{3(2\vec{a}+4\vec{b})+(2\vec{a}-8\vec{b})}{4}=2\vec{a}+\vec{b}.$$

$$\overrightarrow{OD}=\frac{3\overrightarrow{OA}-\overrightarrow{OB}}{3-1}=\frac{3(2\vec{a}+4\vec{b})-(2\vec{a}-8\vec{b})}{2}=2\vec{a}+10\vec{b}.$$

Let us recall the definition that the line joining a vertex of a triangle with the midpoint of its opposite side is called a median. The centroid divides the median from vertex to the midpoint of the opposite side internally in the ratio 2:1.

Theorem 8.3

The medians of a triangle are concurrent.

Proof

Let ABC be a triangle and let D, E, F be the mid points of its sides BC, CA and AB respectively. We have to prove that the medians AD, BE, CF are concurrent.

Let O be the origin and $\vec{a},\vec{b},\vec{c}$ be the position vectors of A, B, and C respectively.

The position vectors of D, E, and F are respectively

$$\frac{\vec{b}+\vec{c}}{2},\frac{\vec{c}+\vec{a}}{2},\frac{\vec{a}+\vec{b}}{2}.$$

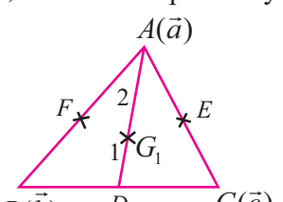


Fig. 8.27

Let G_1 be the point on AD dividing it internally in the ratio 2 : 1

Therefore, position vector of $G_1 = \frac{1\overrightarrow{OA}+2\overrightarrow{OD}}{1+2}$

$$\overrightarrow{OG_1}=\frac{1\vec{a}+2\left(\frac{\vec{b}+\vec{c}}{2}\right)}{3}=\frac{\vec{a}+\vec{b}+\vec{c}}{3} \quad (1)$$

Let G_2 be the point on BE dividing it internally in the ratio 2 : 1

Therefore, $\overrightarrow{OG_2} = \frac{1\overrightarrow{OB}+2\overrightarrow{OE}}{1+2}$

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